

Week 1 — Lecture notes

The core: exponentials, Euler's formula and the complex plane

The figures below are snapshots of interactive widgets. The live versions, where you can move the sliders yourself, are at tchaffey.com/elec2302.

Signals & Systems lives at the intersection of **differential equations**, **linear algebra**, **complex numbers** and **trigonometry**.

(a) The exponential function

For $z \in \mathbb{C}$, define the exponential by its power series:

$$e^z := 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

Properties. First, $e^0 = 1$ (straight from the definition) (1)

$$\begin{aligned} \frac{d}{dz} e^z &= 0 + 1 + \frac{2z}{2!} + \frac{3z^2}{3!} + \dots \\ &= 1 + z + \frac{z^2}{2!} + \dots \\ &= e^z \end{aligned} \quad (2)$$

Combining the two properties:

$$\boxed{f(z) = f'(z), \quad f(0) = 1}$$

It turns out $f(z) = e^z$ is the unique solution to this differential equation, and in fact the differential equation can be used as an alternative definition.

Extension of (2). By the chain rule,

$$\frac{d}{dt} e^{zt} = 0 + z + \frac{2t z^2}{2!} + \frac{3t^2 z^3}{3!} + \dots = z e^{zt}.$$

(b) Euler's formula

Let's now represent sinusoids as complex numbers. Let

$$g(\theta) = \frac{\cos \theta + j \sin \theta}{e^{j\theta}}.$$

Then

$$\begin{aligned} g'(\theta) &= (-\sin \theta + j \cos \theta) e^{-j\theta} - (\cos \theta + j \sin \theta) j e^{-j\theta} \\ &= (-\sin \theta + j \cos \theta) e^{-j\theta} - (-\sin \theta + j \cos \theta) e^{-j\theta} \\ &= 0. \end{aligned}$$

So $g(\theta) = C$ (a constant):

$$C e^{j\theta} = \cos \theta + j \sin \theta.$$

Set $\theta = 0$. Then $C = 1$, so we have

$$e^{j\theta} = \cos \theta + j \sin \theta$$

What does this mean graphically? $e^{j\theta}$ is the unit vector at angle θ ; its real part is $\cos \theta$ and its imaginary part is $\sin \theta$. Advance θ and the vector rotates, its shadow tracing a sinusoid.

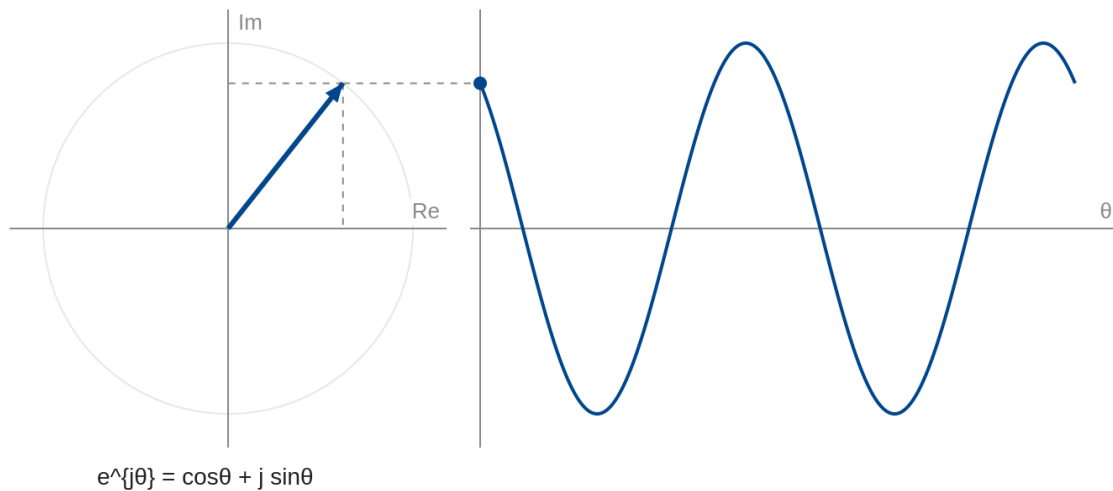


Figure 1: The unit vector $e^{j\theta}$ and its imaginary part unrolling into a sinusoid.

In particular: $e^{j\pi/2} = j$, $e^{j\pi} = -1$.

Manipulating sinusoids using complex numbers

Why use complex numbers to represent sinusoids? Because it makes the algebra easier. For example,

$$x_1(t) = 3 \cos \omega t, \quad x_2(t) = 4 \cos(\omega t + \frac{\pi}{2}), \quad x_1(t) + x_2(t) = ?$$

The direct way uses the identity $\cos(\theta + \frac{\pi}{2}) = -\sin \theta$, so $x_2(t) = -4 \sin \omega t$, giving

$$3 \cos \omega t - 4 \sin \omega t \equiv A \cos(\omega t + \varphi) \quad \longrightarrow \quad \text{match coefficients ...}$$

With complex numbers:

$$\begin{aligned} x_1(t) &= \text{Re}(3 e^{j\omega t}) \\ x_2(t) &= \text{Re}(4 e^{j\omega t + j\pi/2}) = \text{Re}(4 e^{j\pi/2} e^{j\omega t}) = \text{Re}(4j e^{j\omega t}) \\ x_1(t) + x_2(t) &= \text{Re}(3 e^{j\omega t} + 4j e^{j\omega t}) \\ &= \text{Re}((3 + 4j) e^{j\omega t}) \\ &= \text{Re}(5 e^{j0.927} e^{j\omega t}) = 5 \cos(\omega t + 0.927). \end{aligned}$$

The triangle $3 + 4j = 5 \angle 53.1^\circ$ rotates.

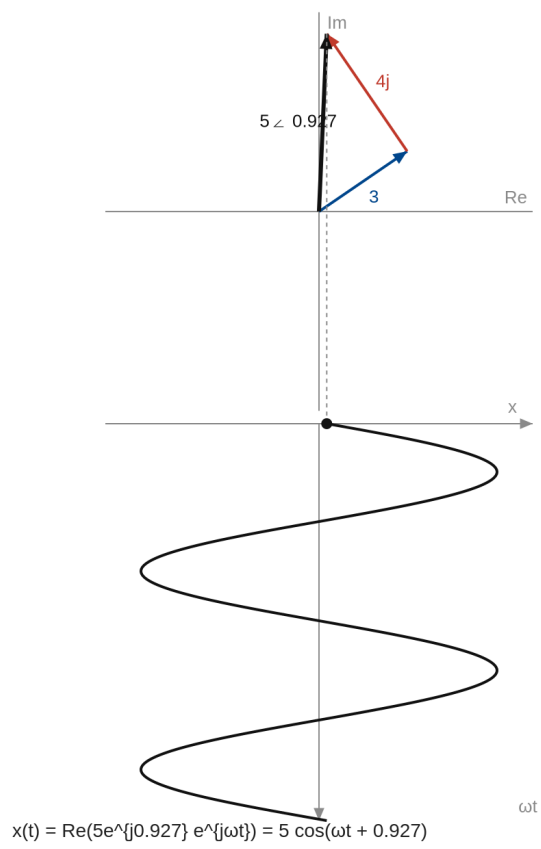


Figure 2: Adding 3 and $4j$ tip to tail. The resultant $5\angle 0.927$ rotates, and its real part traces $5 \cos(\omega t + 0.927)$.

(c) Euler's formula & differential equations

Another differential equation:

$$\frac{d}{dt} x(t) = j x(t) \quad (3)$$

We've already seen $\frac{d}{dt} e^{zt} = z e^{zt}$. Set $z = j$:

$$\frac{d}{dt} e^{jt} = j e^{jt}.$$

This solves (3). What does it look like in $\mathbb{C}$, and what does the real part look like? What about other choices of z ?

$$z = -1 : \quad \frac{d}{dt} e^{-t} = -e^{-t}. \quad \text{No imaginary component.}$$

$$z = -1 + j : \quad \frac{d}{dt} e^{(-1+j)t} = (-1 + j) e^{(-1+j)t}.$$

The widget below plots the trajectory of e^{zt} for several values of z .

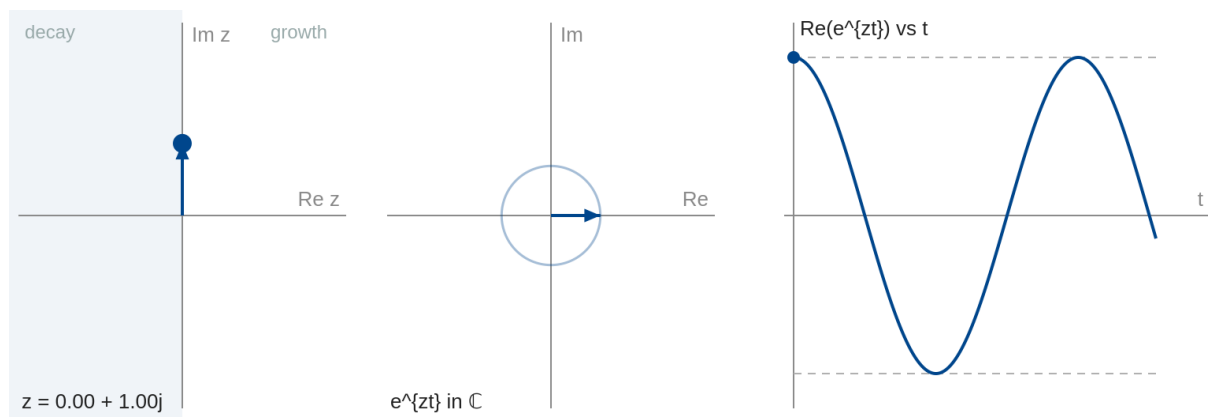


Figure 3: Left: the position of z . Middle: the trajectory of e^{zt} in $\mathbb{C}$. Right: its real part against t . Shown for $z = j$, giving pure oscillation.

The position of z determines the behaviour: oscillations, decay, growth.

(d) What has this got to do with electrical engineering?

Example: an RLC circuit (the **system**) with a sinusoidal voltage input (the **signal**).

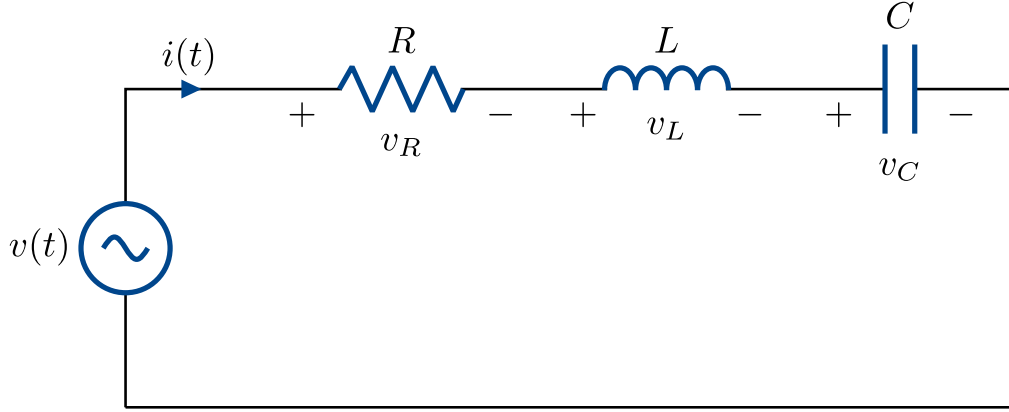


Figure 4: Series RLC circuit.

KVL around the loop:

$$v(t) = v_R + v_L + v_C$$

$$v(t) = R i(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau \quad (4)$$

with the lower limit taken to $-\infty$ for the steady-state response. Let $v(t) = V \cos \omega t = \text{Re}(V e^{j\omega t})$ and look for a current of the same form, $i(t) = \text{Re}(I e^{j\omega t})$.

Substitute in (4):

$$\begin{aligned} V e^{j\omega t} &= R I e^{j\omega t} + L \frac{d}{dt} I e^{j\omega t} + \frac{1}{C} \int I e^{j\omega \tau} d\tau \\ &= R I e^{j\omega t} + j\omega L I e^{j\omega t} + \frac{I}{j\omega C} e^{j\omega t} + \frac{K}{C}, \end{aligned}$$

where K is a constant of integration. Matching coefficients of $e^{j\omega t}$, we find $K = 0$ (K represents a residual charge on the capacitor, which a sinusoidal voltage is not able to maintain at steady state).

Divide through by $e^{j\omega t}$ (never zero):

$$V = I \underbrace{\left(R + j\omega L + \frac{1}{j\omega C} \right)}_{\text{impedance } Z(j\omega)} \implies I = \frac{V}{Z}.$$

Suppose $R = 30 \Omega$ and $\omega L - \frac{1}{\omega C} = 40 \Omega$. Then

$$Z = 30 + 40j = 50 e^{j0.927}, \quad I = \frac{V}{50 e^{j0.927}},$$

$$i(t) = \text{Re}(I e^{j\omega t}) = \frac{V}{50} \cos(\omega t - 0.927).$$

The widget below plots input and output sinusoids for several values of the reactance Z . For some values, the output lags the input, while for others, the output leads.

The RLC circuit maps a sinusoid to a sinusoid with the **same frequency**. This is the foundation of the rest of the course.

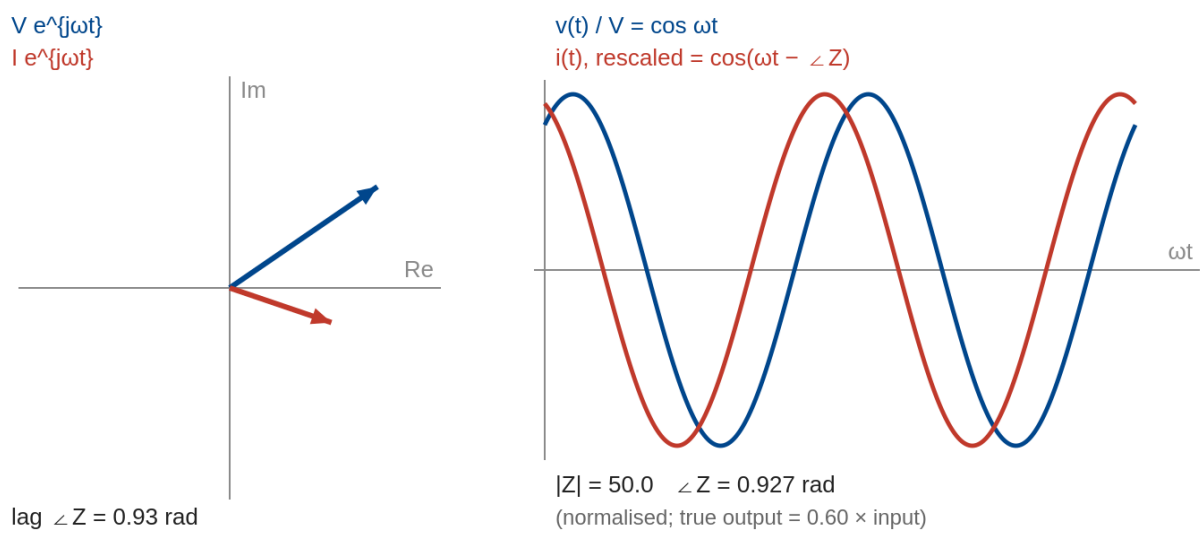


Figure 5: Input and output phasors (left) and waveforms (right), at $R = 30 \Omega$ and $\omega L - 1/\omega C = 40 \Omega$, so $|Z| = 50 \Omega$ and $\angle Z = 0.927$. Amplitudes are normalised here; the live version can show the output at its true size, $R/|Z| = 0.60$ of the input.